Project Feature Guide

AUTOSAR HLD Document Analysis Assistant

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1. Executive Overview

Designed for automotive systems engineers and software architects, the **AUTOSAR HLD Document Analysis Assistant** bridges the gap between unstructured 300-page PDF High-Level Design specifications and actionable architectural intelligence. Using local AI (Ollama + ChromaDB), it provides instant semantic search, component tracing, and verification reports while maintaining human-in-the-loop governance.

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2. Core Feature Catalogue

Feature Name	Business Purpose	User Interaction	Implementation Status
HLD Document Ingestion	Upload and parse complex PDF architectures	Drag-and-drop PDF in Streamlit UI	Implemented
OCR Fallback Processing	Extract text from scanned legacy diagrams/pages	Automatic detection & Tesseract pass	Partially Implemented
Semantic RAG Search	Ask natural-language questions about signals/ports	Chat interface with source citations	Implemented
Entity Extraction	Map components, interfaces, and signals	Structured tabular view in Streamlit	Partially Implemented
Version Comparison	Detect architectural changes across revisions	Side-by-side diff report	Partially Implemented
Validation Checks	Identify unassigned ports or missing dependencies	Automated heuristic warning flags	Partially Implemented

| **Project Isolation** | Separate data across different vehicle programs | Project selector dropdown | Planned (UI only) |

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3. End-to-End Workflow Example

1. **Upload:** Engineer uploads `Brake_System_HLD_v2.1.pdf` via the Streamlit interface.

2. **Ingestion & Parsing:** FastAPI extracts headings, tables, and text chunks.

3. **Indexing:** Embeddings are generated via Sentence Transformers and stored in ChromaDB.

4. **Querying:** Engineer asks: *"Identify the port receiving `VehicleSpeed`."*

5. **Grounded Response:** The system retrieves relevant chunks, queries Ollama (`llama3.2`), and returns:

   > *`VehicleSpeed` is received via port `P_SpeedReceiver` on component `SpeedProcessingSWC` (Source: Section 4.2, Page 28).*